

Tutorial 7 - Arrays

1. Write a C++ program to accept 10 float values using an array and print the numbers.

Sample output

```
Enter 10 numbers
1.5
2.5
3.5
4.5
5.5
6.5
7.5
8.5
9.5
10.5
The numbers are
1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5 10.5
```

2. Write a C++ program to accept 5 characters using an array and print the array in reverse order.

Sample output

```
Enter 5 characters
a
b
c
d
e
characters in reverse order
e d c b a
```

3. Write a C++ program to accept 10 elements of an array and sort the numbers in ascending order and print the numbers.

(logic: compare first two numbers, swap if number in index [1] is larger than index [0]. Repeat the steps index [1] and [2], then [2] and [3]

```
//user entered 6 2 4 8 1
//swap code
temp = arr[j]; //temp=2
arr[j] = arr[i]; //j=6
arr[i] = temp; //i=2 and so on)
```

Sample output

```
Enter 10 numbers
51
5
84
6
25
7
93
41
12
60
Numbers sorted in ascending order
5  6  7  12  25  41  51  60  84  93
```

4. Write a C++ program that accepts 5 integers from the user, stores them in an array, and then prints the following:

- a) The elements of the array.
- b) The sum of all the elements.
- c) The largest number in the array.

(logic: Initialize variable named largest with first element of array.

- o For each element in the array:
- o Compare largest with this element.

If largest is greater than this element, then update largest with the element.

largest contains the largest number in the given integer array)

Sample output

```
Enter 5 integers:
10
3
45
7
22

Array elements: 10 3 45 7 22
Sum of array elements: 87
Largest element: 45
```

5. Write a C++ program that takes two 2x2 matrices as input from the user and then prints the following:

- a) The two matrices itself.
- b) The sum of the two matrices.

Sample output

Enter elements for the first 2x2 matrix:

Element at [0][0]: 1

Element at [0][1]: 1

Element at [1][0]: 1

Element at [1][1]: 1

Enter elements for the second 2x2 matrix:

Element at [0][0]: 2

Element at [0][1]: 2

Element at [1][0]: 2

Element at [1][1]: 2

First Matrix:

1 1

1 1

Second Matrix:

2 2

2 2

Sum of the two matrices:

3 3

3 3